

José Hiram Soltren

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SYNOPSIS

Staff Software/Hardware Engineer: from Silicon through Display to Web.
Seeking Hardware Integration, Firmware, Scientific Computing, Infra roles.
Flight Instructor - CFI, CFII (Instrument), MEI (Multi Engine), AIGI (Ground)
Expert in C, C++, Systems, Linux, device drivers, hardware accelerators.

EXPERIENCE

Connectivity Engineer ForeFlight Jan 2024 - Nov 2024

- Designed and implemented a high-performance UDP-based protocol and API for generic third-party avionics integration, successfully adopted in production by external partners (e.g., uAvionix).
- Engineered a highly optimized payload structure utilizing a novel 4-bit integer format to encode ARINC 424 flight plans, maximizing data transfer efficiency and reliability.
- Resolved a critical, FAA-escalated weather parsing outage by rapidly debugging and fixing a decade-old database query defect within one week of taking ownership, restoring safe flight operations for offshore helicopter fleets.
- Spearheaded the technical infrastructure for an on-site avionics lab, physically integrating hardware from multiple manufacturers to rapidly diagnose and deploy fixes for complex flight plan and data transfer bugs.

Technical Lead, Build and Release Roku Jan 2023 - Sep 2023

- Architected self-service CI/CD infrastructure and implemented pre-merge CI, successfully reducing support tickets by 30% and presenting findings at IEEE ICST 2023.

Senior Software Engineer, Roku Players Roku Jun 2020 - Dec 2022

- Architected and delivered high-performance, multi-threaded Linux applications for RokuOS firmware using C++20, ensuring ultra-reliable video decode and multimedia playback.
- Spearheaded Apple AirPlay and YouTube certification across all HDMI player platforms, designing fundamental system APIs and refactoring legacy code to satisfy strict third-party partner requirements.
- Served as Firmware Lead for the Roku Streaming Stick 4K (20M+ units); embedded with the hardware team to root-cause and resolve a critical device brownout defect by debugging and fixing an errant SoC power configuration.
- Engineered a novel, system-level debug flow—synchronizing multi-source logs, timestamps, and video frame captures alongside gdb and git—to methodically resolve complex, multi-process software defects.
- Leveraged a highly detail-oriented approach within the internal alpha program to identify and resolve critical video quality bugs across middleware layers, bridging RokuOS software and SoC vendor code.

Deep Learning Software Engineer Nervana / Intel Corp. Aug 2017 - Feb 2019

- Firmware development in C++ for deep learning training hardware accelerator.
- Focus on hardware and software to run GEMM and convolution as fast as possible.
- Designed an algorithm for all-to-all data transfers on our hardware that beat the Bruck Algorithm, the fastest general purpose algorithm.

Software Engineer Cloudera, Inc.

May 2016 - Aug 2017

- Engineered distributed fault tolerance and advanced parallelism mechanisms directly within Apache Spark internals (Scala/JVM), presenting architectural findings at Spark Summit.
- Developed robust tooling for cluster analysis and automated fault detection, identifying bottlenecks and mitigating failure modes in large-scale concurrent environments.

Senior System Software Engineer NVIDIA Corporation May 2011 - Apr 2016

- Firmware and driver development for 3D graphics accelerators.
- Lead engineer for video decode and post-processing, including hardware validation.
- Bug fix and feature work in OpenGL and Vulkan.

ADDITIONAL
ACTIVITIES

Commercial Pilot & Flight Instructor Soltren Consulting Apr 2024 - present

- Systems Mastery: Logged over 1,250 flight hours operating safety-critical aerospace systems, including 500+ hours in high-performance aircraft and extensive expertise in complex Garmin avionics suites.
- Technical Excellence: Demonstrated extreme attention to detail and a strong track record of success across numerous rigorous FAA technical exams and practical flight evaluations.
- Mentorship & Leadership: Delivered 670+ hours of advanced instruction, specializing in mentoring commercial pilots through complex aircraft transitions and instructor ratings.
- Operational Execution: Serve as a Volunteer Mission Pilot and Instructor for the Civil Air Patrol, executing high-stakes flight profiles under stringent operational and compliance guidelines.

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Master of Engineering and S.B., Electrical Engineering and Computer Science

Grad level operating systems course work. Build a kernel and hypervisor in C.

Lab assistant for undergraduate microcontroller/firmware course.

CITIZENSHIP

United States